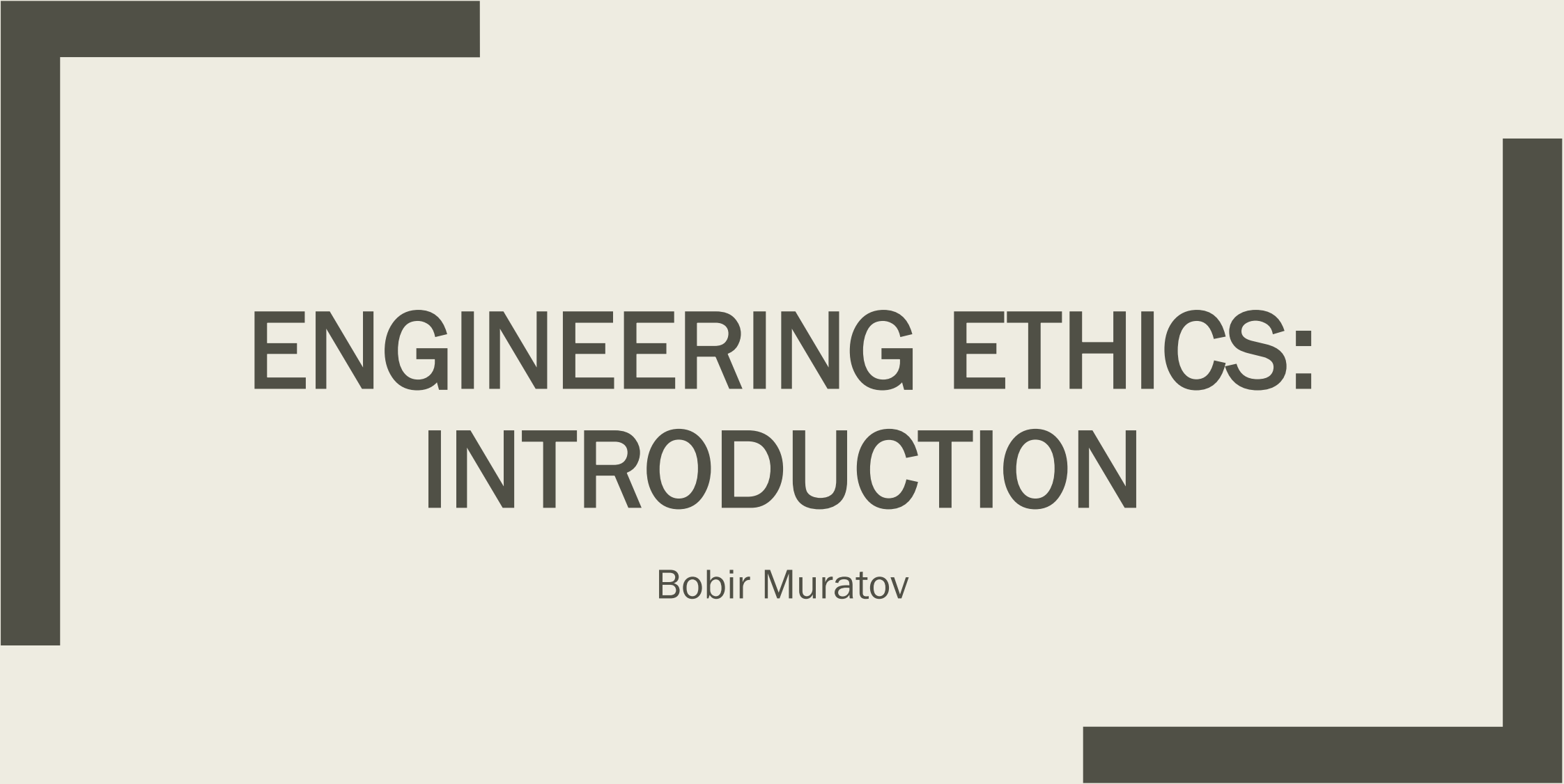


A dark gray L-shaped frame composed of two thick bars. One bar is vertical on the left side, and the other is horizontal at the top, meeting at a right angle in the upper-left corner. Another similar L-shaped bar is positioned on the right side, with a vertical bar and a horizontal bar meeting at a right angle in the lower-right corner.

ENGINEERING ETHICS: INTRODUCTION

Bobir Muratov



Tesla Case

- Where should moral responsibility and legal liability lie in this case?
- How realistic is it to install an autopilot system and then tell the driver she must be able to take over at any time?
- How safe are these cars?
- What kinds of information should be given to drivers before they purchase or use these vehicles?
- How should the potential problems of hacking and terrorism be handled?
- What about the potential loss of driving-related jobs?
- Who is responsible?

What is Professionalism?

Professionalism

- expertise in a certain area (accounting, law, medicine, engineering, etc.)
- adherence to moral guidelines, usually laid out in a formal code of ethics.

What is a Profession?

Profession is

- Sociological Account
- Social Contract Account
- Public element

BOX 1.2 Characteristics of a Profession

1. Extensive period of training of an intellectual character, usually obtained at a college or university.
2. Possessing knowledge and skills vital to the well-being of the larger society.
3. A monopoly or near-monopoly on the provision of professional services, and considerable control over professional education and the standards for admission into the profession.
4. An unusual degree of autonomy in the workplace.
5. A claim to be regulated by ethical standards, usually embodied in a code of ethics, that promotes the good of the public.³

Is Engineering a Profession?

Should Engineers Serve to Public
Good?

What is the Public Good?

Public Good is

- engineers should hold paramount the safety, health, and welfare of the public
- engineering has a direct and vital impact on the quality of life for all people
- contribute to society and human well-being
- engineers should utilize their knowledge and skill for the enhancement of human welfare and the environment
- promoting the well-being of the public is the primary responsibility of the engineering profession

What is Well-Being?

Well-Being is

- Capabilities Approach by Amartya Sen and Martha Nussbaum:
- “certain living conditions are essential to a preferred quality in our own lives”
- having food, shelter, and water, having satisfying human relationships (communication, the Internet), having free movement and expression (highways, air travel, the Internet, telephone, etc.), and having a satisfactory relationship to the natural world (environmental preservation)

The Types of Actions that Guide the Engineers' Ethics

- Prohibited actions
- Preventing harm
- Promoting well-being

Prohibited Actions vs. Preventing Harm

- Prevention of harm usually involves:
- identifying and disclosing potential harms
- attempting to prevent them

BOX 1.4 Examples of Prohibited Actions from the NSPE Code

- Do not reveal privileged information (II,1,c)
- Do not associate with dishonest professionals (II,1,d)
- Do not aid the unlawful practice of engineering (II,1,e)
- Do not accept compensation from two parties on the same project (II,4,b)
- Do not participate in governmental decisions related to your own work (II,4,d)
- Do not solicit work from a governmental body on which a member of your firm has a position (II,4,e)
- Do not falsify your qualifications (II,5,a)
- Do not give bribes (II,5,b)
- Do not be influenced by conflicting interests (III,5)
- Do not unjustly injure the reputation of another engineer (III,7)

Promoting Well-Being: Aspirational Ethics

BOX 1.5 NAE's List of the Greatest Engineering Achievements of the Twentieth Century

- Electrification
- The automobile
- The airplane
- Water supply and distribution systems
- Electronics (vacuum tubes, transistors, etc.)
- Radio and television
- Agricultural mechanization
- Computers
- Telephones
- Air-conditioning and refrigeration
- Highways
- Spacecraft
- The Internet
- Imaging (especially in medicine)
- Household appliances
- Health technologies
- Petroleum and petroleum technologies
- Laser and fiber optics
- Nuclear technologies
- High-performance materials

BOX 1.6 NAE's List of the Grand Challenges for the Twenty-First Century

- Make solar energy economical
- Provide energy from fusion
- Develop carbon sequestration methods
- Manage the nitrogen cycle
- Provide access to clean water
- Restore and improve urban infrastructure
- Advance health informatics
- Engineer better medicines
- Reverse-engineer the brain
- Prevent nuclear terror
- Secure cyberspace
- Enhance virtual reality
- Advance personalized learning
- Engineer the tools of scientific discovery

Social Context of Technology

- engineering is a type of social experimentation
- technology affects society
- the social context influences the development of technology

Critical Attitude toward Technology

- technological determinism
- technological optimism
- technological pessimism

- where is the truth?

Class Administration & Syllabus

Communication

- Office: B409
- Office hours: Wednesdays 15:30-17:30
- E-class - message
- Email: b.muratov@inha.uz
- Curator
- Telegram?
- LinkedIn?

Class Policy

- Attendance (F for missing 25%)
- Max. 10 minutes can be late
- No phones and laptops
- Break time in the middle of class

Grading

- Attendance (20%)
- Participation (20%)
- Paper presentation (30%)
- Midterm (30%)

Textbook

Sixth Edition

Concepts and Cases

ENGINEERING ETHICS

CHARLES E. HARRIS, JR.

MICHAEL S. PRITCHARD

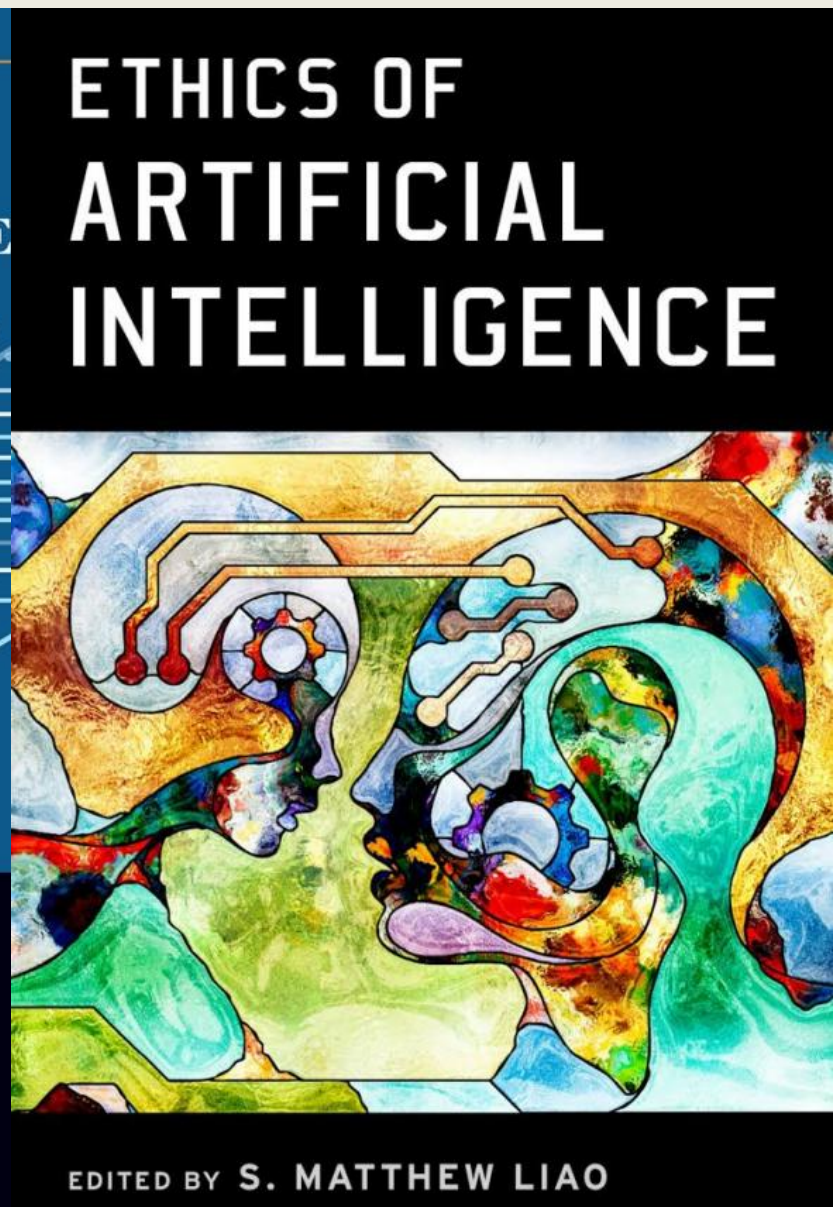
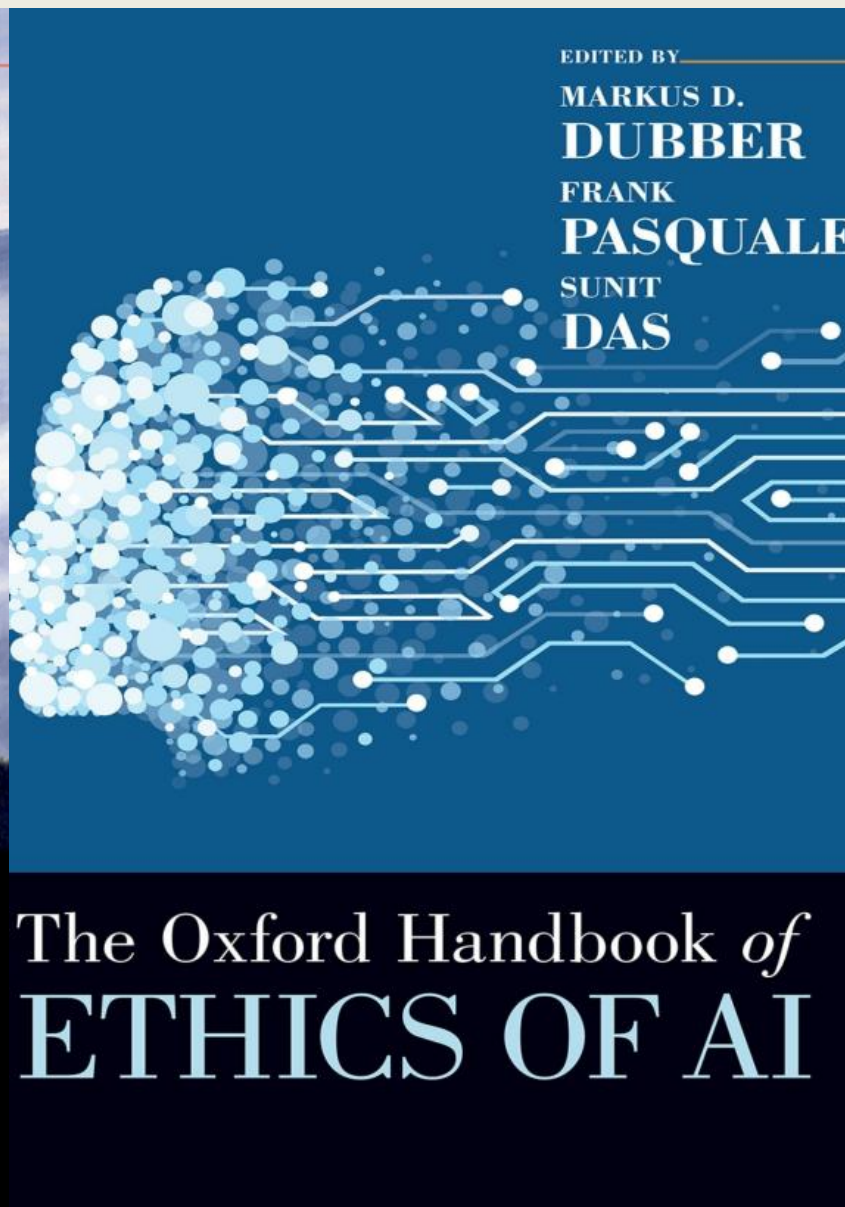
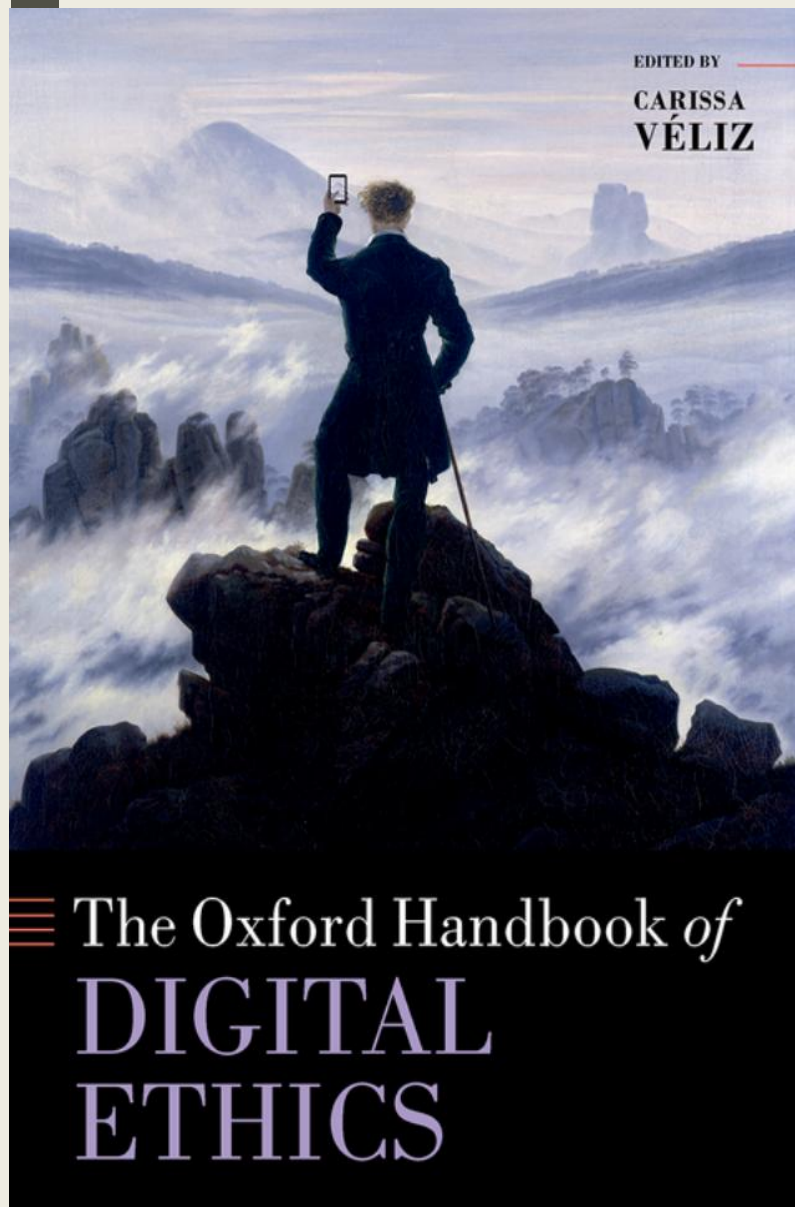
RAY W. JAMES, P.E.

ELAINE E. ENGLEHARDT

MICHAEL J. RABINS

Topics

- Engineers: Professionals for the Human Good
- A Practical Ethics Toolkit
- Responsibility in Engineering
- Engineers in Organizations
- Trust and Reliability
- The Engineer's Responsibility to Assess and Manage Risk
- Engineering and the Environment
- Engineering in the Global Context
- New Horizons in Engineering



Paper presentation

- Teams of 5 students
- 1 team member fills the survey – list of members and paper
- Each member of the team should submit the PPT file
- Presentation content - based on the sections of papers and beyond
- Time limit: 20 (min.) – 25 (max.) minutes.
- Grading criteria:
 - ✓ Participation of all members
 - ✓ Presentation skills
 - ✓ Depth of content
 - ✓ Quality of discussion
 - ✓ Others
- Deadline/start: 2nd week
- 2-3 teams present each week